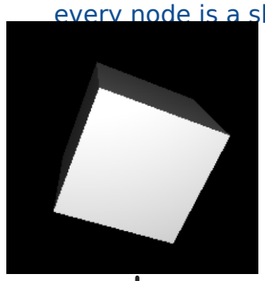
NeST SO(3) decode tree on one real image: the beam-MAP path through the Hopf--HEALPix tree

every node is a shaded 3D globe of the Hopf--HEALPix S^2 base; filled HEALPix cells are the density (alpha = probability), hue = the Hopf roll fibre; each branch probability is the model's 8-way softmax

gold = the beam-MAP root → leaf path (branch width ∝ probability); the chosen bin+fibre refines ×8 (pixel ×4, roll ×2) each level

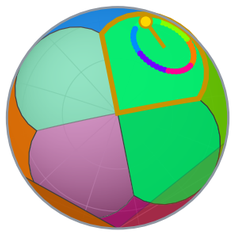


actual input
(cube mask)

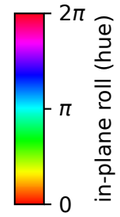
ViT-B/16 encoder

tokens condition every node

root $p_0(c_0 \mid x)$
72-way · full density

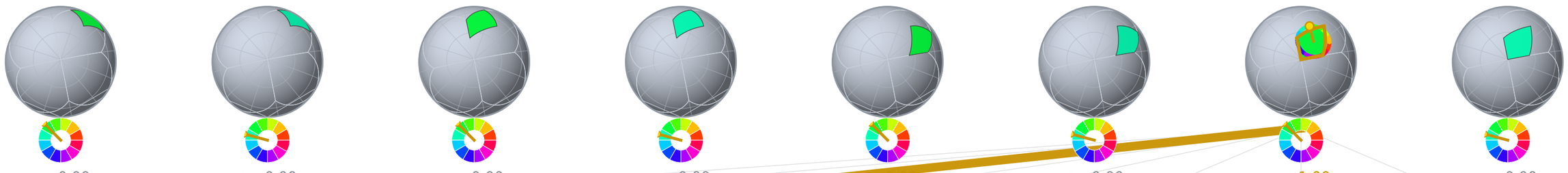






$\ell = 1$

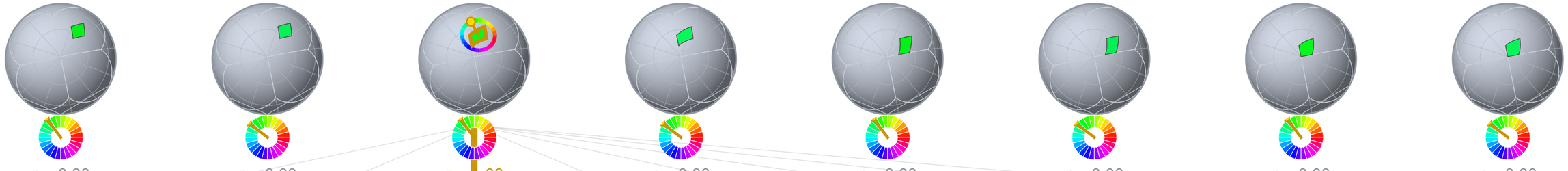
8-way $p(c_1 \mid \cdot)$ nside 2



$p = 0.00$ $p = 0.00$ $p = 0.00$ $p = 0.00$ $p = 0.00$ $p = 0.00$ $p = 1.00$ $p = 0.00$

$\ell = 2$

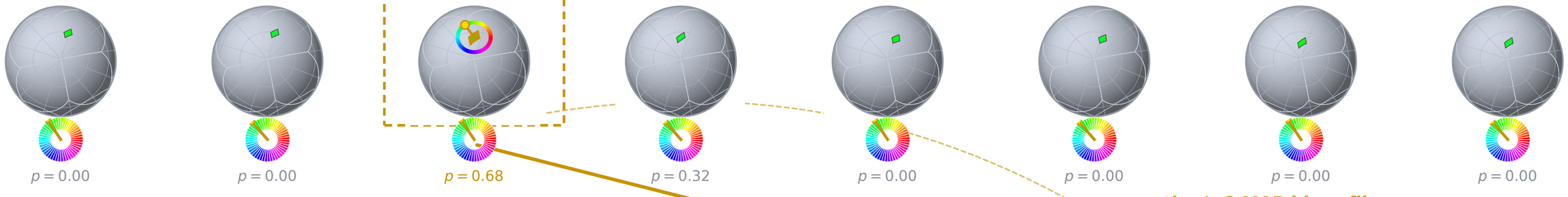
8-way $p(c_2 \mid \cdot)$ nside 4



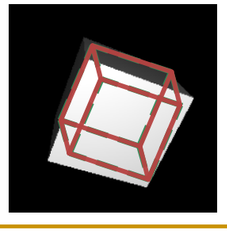
$p = 0.00$ $p = 0.00$ $p = 1.00$ $p = 0.00$ $p = 0.00$ $p = 0.00$ $p = 0.00$ $p = 0.00$

$\ell = 3$

8-way $p(c_3 \mid \cdot)$ nside 8



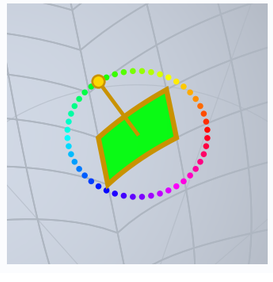
$p = 0.00$ $p = 0.00$ $p = 0.68$ $p = 0.32$ $p = 0.00$ $p = 0.00$ $p = 0.00$ $p = 0.00$



nearest sym. GT — NeST MAP

leaf node =
output orientation R
geodesic err (min over cube symmetry) $= 0.9^\circ$

zoom: the ℓ=3 MAP bin + fibre



nside 8
pix 53
roll 16/48
$p = 0.68$